Problem Set 1

Detailed solutions

Week 1: Fundamentals of Quantum Computing

This manual covers **Questions 1–4 and 5.1–5.3**, with the same subquestion numbering as the problem set. Each solution includes the calculation or reasoning behind the result.

Conventions

The computational basis is orthonormal: $\langle j \mid k \rangle = \delta_{jk}$. Column-vector entries follow the basis order. For multiple qubits we use

$$(|00\rangle, |01\rangle, |10\rangle, |11\rangle), \qquad (|000\rangle, |001\rangle, \ldots, |111\rangle).$$

The **leftmost bit corresponds to the bottom circuit wire**. Qubits are labelled downward from the top: q_0 is the top wire, q_1 is the next wire down, and so on. For two qubits,

$$|ab\rangle = |a\rangle_{q_1} \otimes |b\rangle_{q_0}, \qquad \underbrace{a}_{\text{bottom wire}} \quad \underbrace{b}_{\text{top wire}}.$$

Thus $A \otimes B$ applies A to the bottom wire q_1 and B to the top wire q_0. For three qubits, the left, middle and right bits correspond to q_2 (bottom), q_1 (middle) and q_0 (top), respectively. Vector entries and matrix rows and columns follow the displayed lexicographic basis order.

Circuits run from left to right; in a matrix product, the rightmost operator acts first. Unless stated otherwise, measurement means measurement in the computational basis.

Three rules used throughout. If $|u\rangle = \sum_j u_j |j\rangle$ and $|v\rangle = \sum_j v_j |j\rangle$, then

$$\langle u| = \sum_j u_j^* \langle j|, \qquad \langle u \mid v \rangle = \sum_j u_j^* v_j, \qquad (|u\rangle\langle v|)_{jk} = u_j v_k^*.$$

A bra is a *conjugate transpose*, an inner product is a scalar, and an outer product is a matrix. For a normalized quantum state, the probability of outcome j is $|u_j|^2$.

In Question 1.11, "classical state" is interpreted as a **classical probability vector**: its entries are real, nonnegative, and sum to one. This differs from the normalization rule for a pure quantum state, whose *squared magnitudes* sum to one.

Question 1

Two-level system: $\Sigma = \{0, 1\}$

1.1 Column vectors to Dirac notation

Because $|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$ and $|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$, the conversion rule is

$$\begin{bmatrix}a\\b\end{bmatrix} = a|0\rangle + b|1\rangle.$$

Reading off the entries, the seven answers, in the order given, are

(i) $\begin{bmatrix}1\\0\end{bmatrix} = |0\rangle,$ (ii) $\begin{bmatrix}0\\1\end{bmatrix} = |1\rangle,$

(iii) $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\\mathrm{i}\end{bmatrix} = \frac{|0\rangle + \mathrm{i}|1\rangle}{\sqrt{2}} = |+i\rangle,$ (iv) $\frac{1}{\sqrt{5}}\begin{bmatrix}2\\1\end{bmatrix} = \frac{2|0\rangle + |1\rangle}{\sqrt{5}},$

(v) $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle,$ (vi) $\frac{1}{\sqrt{10}}\begin{bmatrix}3\\1\end{bmatrix} = \frac{3|0\rangle + |1\rangle}{\sqrt{10}},$

(vii) $\frac{1}{\sqrt{2}}\begin{bmatrix}\mathrm{i}\\1\end{bmatrix} = \frac{\mathrm{i}|0\rangle + |1\rangle}{\sqrt{2}} = \mathrm{i}|-i\rangle.$

In (vii), the factor i is a global phase: the displayed vector and $|-i\rangle$ give the same physical state, although they are not the same vector.

1.2 Dirac notation to column vectors

Expand each ket in the computational basis and put its coefficients into a column:

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}, \qquad |+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}, \qquad |-i\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-\mathrm{i}\end{bmatrix},$$

$$|\theta\rangle = \frac{1}{2}\begin{bmatrix}\sqrt{3}\\1\end{bmatrix}, \qquad |\eta\rangle = \frac{1}{\sqrt{5}}\begin{bmatrix}2\\-\mathrm{i}\end{bmatrix}.$$

For $|\xi\rangle$, first substitute the definitions of $|-\rangle$ and $|+\rangle$:

$$\begin{aligned}|\xi\rangle &= \frac{1}{\sqrt{10}}\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} + 3\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\\ &= \frac{4|0\rangle + 2|1\rangle}{\sqrt{20}} = \frac{2|0\rangle + |1\rangle}{\sqrt{5}} = \boxed{\frac{1}{\sqrt{5}}\begin{bmatrix}2\\1\end{bmatrix}}.\end{aligned}$$

This computational-basis form will make the next two parts easier.

1.3 Inner products

The coefficients in the *bra* must be complex conjugated. In particular, $\langle +i| = (\langle 0| - \mathrm{i}\langle 1|)/\sqrt{2}$ and $\langle -i| = (\langle 0| + \mathrm{i}\langle 1|)/\sqrt{2}$.

$$\langle - \mid + \rangle = \frac{1}{2}\begin{bmatrix}1 & -1\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1-1}{2} = \boxed{0},$$

$$\langle + \mid +i \rangle = \frac{1}{2}\begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1\\\mathrm{i}\end{bmatrix} = \boxed{\frac{1+\mathrm{i}}{2}},$$

$$\langle +i \mid + \rangle = \frac{1}{2}\begin{bmatrix}1 & -\mathrm{i}\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \boxed{\frac{1-\mathrm{i}}{2}},$$

$$\langle -i \mid 0 \rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & \mathrm{i}\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \boxed{\frac{1}{\sqrt{2}}},$$

$$\langle + \mid \theta \rangle = \frac{1}{2\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}\sqrt{3}\\1\end{bmatrix} = \boxed{\frac{\sqrt{3}+1}{2\sqrt{2}}},$$

$$\langle \xi \mid \eta \rangle = \frac{1}{5}\begin{bmatrix}2 & 1\end{bmatrix}\begin{bmatrix}2\\-\mathrm{i}\end{bmatrix} = \boxed{\frac{4-\mathrm{i}}{5}}.$$

As a check, $\langle + \mid +i\rangle$ and $\langle +i \mid +\rangle$ are complex conjugates.

1.4 Outer products

For columns $u = \begin{bmatrix}a\\b\end{bmatrix}$ and $v = \begin{bmatrix}c\\d\end{bmatrix}$,

$$uv^\dagger = \begin{bmatrix}a\\b\end{bmatrix}\begin{bmatrix}c^* & d^*\end{bmatrix} = \begin{bmatrix}ac^* & ad^*\\bc^* & bd^*\end{bmatrix}.$$

Therefore

$$|0\rangle\langle 0| = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \qquad |+\rangle\langle +| = \frac{1}{2}\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix},$$

$$|1\rangle\langle +| = \frac{1}{\sqrt{2}}\begin{bmatrix}0 & 0\\1 & 1\end{bmatrix}, \qquad |\eta\rangle\langle\eta| = \frac{1}{5}\begin{bmatrix}2\\-\mathrm{i}\end{bmatrix}\begin{bmatrix}2 & \mathrm{i}\end{bmatrix} = \frac{1}{5}\begin{bmatrix}4 & 2\mathrm{i}\\-2\mathrm{i} & 1\end{bmatrix},$$

$$|\xi\rangle\langle\theta| = \frac{1}{2\sqrt{5}}\begin{bmatrix}2\\1\end{bmatrix}\begin{bmatrix}\sqrt{3} & 1\end{bmatrix} = \frac{1}{2\sqrt{5}}\begin{bmatrix}2\sqrt{3} & 2\\\sqrt{3} & 1\end{bmatrix}.$$

The matrices $|0\rangle\langle 0|$, $|+\rangle\langle +|$ and $|\eta\rangle\langle\eta|$ are projectors: they are Hermitian, have trace one, and square to themselves. An outer product with different kets, such as $|1\rangle\langle +|$, need not be Hermitian.

1.5 Applying X, Y, Z and H

Write $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix}\alpha\\ \beta\end{bmatrix}$. Direct matrix multiplication gives both requested forms:

$$X|\psi\rangle = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \boxed{\begin{bmatrix}\beta\\ \alpha\end{bmatrix}} = \boxed{\beta|0\rangle + \alpha|1\rangle},$$

$$Y|\psi\rangle = \begin{bmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \boxed{\begin{bmatrix}-\mathrm{i}\beta\\ \mathrm{i}\alpha\end{bmatrix}} = \boxed{-\mathrm{i}\beta|0\rangle + \mathrm{i}\alpha|1\rangle},$$

$$Z|\psi\rangle = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \boxed{\begin{bmatrix}\alpha\\ -\beta\end{bmatrix}} = \boxed{\alpha|0\rangle - \beta|1\rangle},$$

$$H|\psi\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \boxed{\frac{1}{\sqrt{2}}\begin{bmatrix}\alpha+\beta\\ \alpha-\beta\end{bmatrix}}$$
$$= \boxed{\frac{\alpha+\beta}{\sqrt{2}}|0\rangle + \frac{\alpha-\beta}{\sqrt{2}}|1\rangle}.$$

1.6 Probability of measuring 0

Apply the Born rule to the coefficient of $|0\rangle$ in each result above:

$$\boxed{\begin{aligned} &P(0 \mid X\psi) = |\beta|^2, \quad P(0 \mid Y\psi) = |-\mathrm{i}\beta|^2 = |\beta|^2,\\ &P(0 \mid Z\psi) = |\alpha|^2, \quad P(0 \mid H\psi) = \frac{|\alpha+\beta|^2}{2}. \end{aligned}}$$

The Hadamard result contains an interference term:

$$\frac{|\alpha+\beta|^2}{2} = \frac{|\alpha|^2 + |\beta|^2 + \alpha\beta^* + \alpha^*\beta}{2} = \frac{1}{2} + \mathrm{Re}(\alpha\beta^*).$$

Thus relative phase can change a measurement probability after H. For example, $H|+\rangle = |0\rangle$ gives probability one, whereas $H|-\rangle = |1\rangle$ gives probability zero.

1.7 Reversing the order of an inner product

Let $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$. Then

$$\langle\psi \mid \phi\rangle = \alpha^*\gamma + \beta^*\delta, \qquad \langle\phi \mid \psi\rangle = \gamma^*\alpha + \delta^*\beta = \big(\langle\psi \mid \phi\rangle\big)^*.$$

Hence $\boxed{\langle\psi \mid \phi\rangle = \big(\langle\phi \mid \psi\rangle\big)^*}$. This is *conjugate symmetry*; the two inner products need not be equal. The identity holds even for vectors that are not normalized.

1.8 Operators written with kets and bras

The matrix $|j\rangle\langle k|$ has a single 1 in row j, column k, with all other entries zero. Thus any operator satisfies $A = \sum_{j,k} A_{jk}|j\rangle\langle k|$. Reading the entries of the four matrices gives

$$\begin{aligned} X &= |0\rangle\langle 1| + |1\rangle\langle 0|, \\ Y &= -\mathrm{i}|0\rangle\langle 1| + \mathrm{i}|1\rangle\langle 0|, \\ Z &= |0\rangle\langle 0| - |1\rangle\langle 1|, \\ H &= \frac{1}{\sqrt{2}}\left(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|\right). \end{aligned}$$

1.9 Unitarity of Y, Z and H

A matrix U is unitary when $U^\dagger U = I$. Each of these three matrices is Hermitian:

$$Y^\dagger = Y, \qquad Z^\dagger = Z, \qquad H^\dagger = H.$$

For Y, conjugating changes the signs of the imaginary entries and transposing interchanges their positions, giving Y again. The other two matrices are real and symmetric. It remains to compute their squares:

$$\begin{aligned} Y^\dagger Y = Y^2 &= \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}\begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix} = \begin{bmatrix} (-\mathrm{i})\mathrm{i} & 0 \\ 0 & \mathrm{i}(-\mathrm{i}) \end{bmatrix} = I, \\ Z^\dagger Z = Z^2 &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I, \\ H^\dagger H = H^2 &= \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I. \end{aligned}$$

Therefore Y, Z and H are all unitary. In particular, each preserves the norm of every state vector.

1.10 The X eigenstates

Use $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$:

$$\begin{aligned} X|+\rangle &= \frac{X|0\rangle + X|1\rangle}{\sqrt{2}} = \frac{|1\rangle + |0\rangle}{\sqrt{2}} = +|+\rangle, \\ X|-\rangle &= \frac{X|0\rangle - X|1\rangle}{\sqrt{2}} = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle. \end{aligned}$$

Thus $|+\rangle$ has eigenvalue $+1$ and $|-\rangle$ has eigenvalue -1. They also form an orthonormal basis, as $\langle - \mid + \rangle = 0$ and each has norm one.

1.11 Quantum state, classical state, or neither?

We test each vector *as written*. Multiplying it by a normalization factor would give a different vector and would change the question.

> A pure quantum state requires $\sum_j |a_j|^2 = 1$. A classical probability vector requires $a_j \in \mathbb{R}$, $a_j \geq 0$, and $\sum_j a_j = 1$.

(i) $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$**.** Its entries are nonnegative and sum to one, so it is a valid **classical probability vector**. It is not a normalized quantum state because

$$\|\psi\|^2 = \left|\frac{1}{2}\right|^2 + \left|\frac{1}{2}\right|^2 = \frac{1}{2} \neq 1.$$

(ii) $|\psi\rangle = \frac{1}{3}|0\rangle - \frac{2\mathrm{i}}{3}|1\rangle$**.** The second entry is not a real, nonnegative probability. Also,

$$\|\psi\|^2 = \frac{1}{9} + \frac{4}{9} = \frac{5}{9} \neq 1.$$

The answer is **neither**.

(iii) $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \frac{\sqrt{2}\,\mathrm{i}}{\sqrt{3}}|1\rangle$**.** Here

$$\|\psi\|^2 = \frac{1}{3} + \frac{2}{3} = 1.$$

This is a valid **quantum state**. Its amplitudes are not a classical probability vector. Its computational-basis measurement probabilities are $1/3$ and $2/3$.

(iv) $|\psi\rangle = \sqrt{2}|0\rangle - |+\rangle$**.** The two terms are not orthogonal, so we first expand in the computational basis:

$$\sqrt{2}|0\rangle - \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

This is a valid **quantum state**. The negative amplitude prevents the displayed vector from being a classical probability vector.

Vector	Squared norm	Classification
(i)	$1/2$	Classical probability vector
(ii)	$5/9$	Neither
(iii)	1	Quantum state
(iv)	1	Quantum state

Conceptual distinction. The classical distribution $(1/2, 1/2)$ and the pure state $|+\rangle$ give the same computational-basis probabilities, but they are different physical preparations. If the classical distribution is encoded as a quantum mixture, it has density matrix $\frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$; the pure state $|+\rangle$ has the off-diagonal terms shown in Question 1.4.

Question 2

Five-level system: $\Sigma = \{0, 1, 2, 3, 4\}$

2.1 Column vectors to Dirac notation

The entry in position $k+1$ is the coefficient of $|k\rangle$:

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \sum_{k=0}^{4} a_k |k\rangle.$$

Reading the six vectors in the given order gives

$$\begin{aligned}
&\text{(i)} && |0\rangle, \\
&\text{(ii)} && |2\rangle, \\
&\text{(iii)} && \frac{|1\rangle + |3\rangle}{\sqrt{2}}, \\
&\text{(iv)} && \frac{|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle}{\sqrt{5}}, \\
&\text{(v)} && \frac{e^{\mathrm{i}\pi/4}|1\rangle + e^{-\mathrm{i}\pi/4}|2\rangle}{\sqrt{2}}, \\
&\text{(vi)} && \frac{2|0\rangle - |1\rangle + \mathrm{i}|3\rangle}{\sqrt{6}}.
\end{aligned}$$

For example, (vi) is normalized because $(|2|^2 + |-1|^2 + |\mathrm{i}|^2)/6 = (4+1+1)/6 = 1$.

2.2 Dirac notation to column vectors

Using the basis order $(|0\rangle, |1\rangle, |2\rangle, |3\rangle, |4\rangle)$,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad |4\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad |\nu\rangle = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad |\omega\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}.$$

Define $r = e^{2\pi\mathrm{i}/5}$. Then $r^5 = 1$, $r \neq 1$, and $|r| = 1$. The last vector is

$$\boxed{|\mu\rangle = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ r \\ r^2 \\ r^3 \\ r^4 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ e^{2\pi\mathrm{i}/5} \\ e^{4\pi\mathrm{i}/5} \\ e^{6\pi\mathrm{i}/5} \\ e^{8\pi\mathrm{i}/5} \end{bmatrix}.}$$

All five entries of $|\mu\rangle$ have squared magnitude $1/5$.

2.3 Inner products

Orthonormality of the computational basis immediately gives

$$\boxed{\langle 3 \mid 4\rangle = 0}, \qquad \boxed{\langle 2 \mid \nu\rangle = \frac{1}{\sqrt{5}}}.$$

For the remaining two products,

$$\langle \nu \mid \omega\rangle = \frac{1}{\sqrt{10}}[1 \;\; 1 \;\; 1 \;\; 1 \;\; 1] \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} = \frac{1-1}{\sqrt{10}} = \boxed{0},$$

$$\langle \nu \mid \mu\rangle = \frac{1}{5}(1 + r + r^2 + r^3 + r^4) = \boxed{0}.$$

The last sum vanishes by the geometric-series identity:

$$(1-r)(1 + r + r^2 + r^3 + r^4) = 1 - r^5 = 0.$$

Since $r \neq 1$, division by $1 - r$ gives $1 + r + r^2 + r^3 + r^4 = 0$.

2.4 Outer products

(i) $|1\rangle\langle 3|$**.** This operator sends $|3\rangle$ to $|1\rangle$ and sends the other four basis vectors to zero. Its only nonzero entry is in the second row and fourth column:

$$\boxed{|1\rangle\langle 3| = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}.$$

(ii) $|0\rangle\langle\omega|$**.** Since $\langle\omega| = (\langle 1| - \langle 3|)/\sqrt{2}$, only the first row is nonzero:

$$\boxed{|0\rangle\langle\omega| = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}.$$

(iii) $|\mu\rangle\langle\mu|$**.** Because $(r^k)^* = r^{-k}$, the entry with indices $j, k \in \{0, \ldots, 4\}$ is $\frac{1}{5}r^{j-k}$. Reducing powers modulo five gives

$$\boxed{|\mu\rangle\langle\mu| = \frac{1}{5} \begin{bmatrix} 1 & r^4 & r^3 & r^2 & r \\ r & 1 & r^4 & r^3 & r^2 \\ r^2 & r & 1 & r^4 & r^3 \\ r^3 & r^2 & r & 1 & r^4 \\ r^4 & r^3 & r^2 & r & 1 \end{bmatrix}}.$$

As checks, the diagonal sums to one and entry (k, j) is the complex conjugate of entry (j, k).

Question 3
Tensor products

3.1 Column vectors of product states

In lexicographic order, the Kronecker-product rule is

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}.$$

The first factor varies more slowly: the first two entries have bottom-qubit value 0, and the last two have bottom-qubit value 1.

$$\text{(i)} \quad |0\rangle \otimes |+\rangle = \frac{|00\rangle + |01\rangle}{\sqrt{2}} = \boxed{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}},$$

$$\text{(ii)} \quad |+i\rangle \otimes |1\rangle = \frac{|01\rangle + \mathrm{i}|11\rangle}{\sqrt{2}} = \boxed{\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ \mathrm{i} \end{bmatrix}}.$$

(iii) $|+\rangle \otimes |+\rangle \otimes |+i\rangle$**.** The bottom and middle qubits each contribute $1/\sqrt{2}$ to either basis value. The top qubit contributes $1/\sqrt{2}$ for 0 and $\mathrm{i}/\sqrt{2}$ for 1. Thus

$$\begin{aligned} |+\rangle \otimes |+\rangle \otimes |+i\rangle = \frac{1}{2\sqrt{2}} \big(& |000\rangle + \mathrm{i}|001\rangle + |010\rangle + \mathrm{i}|011\rangle \\ & + |100\rangle + \mathrm{i}|101\rangle + |110\rangle + \mathrm{i}|111\rangle \big) \end{aligned}$$

$$= \boxed{\frac{1}{2\sqrt{2}} \begin{bmatrix} 1 \\ \mathrm{i} \\ 1 \\ \mathrm{i} \\ 1 \\ \mathrm{i} \\ 1 \\ \mathrm{i} \end{bmatrix}}.$$

All eight basis outcomes have probability $1/8$.

3.2 The vector and projector for $|\chi\rangle$

The coefficients of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ are $0, 1/2, -\sqrt{3}/2, 0$, so

$$\boxed{|\chi\rangle = \frac{1}{2} \begin{bmatrix} 0 \\ 1 \\ -\sqrt{3} \\ 0 \end{bmatrix}}, \qquad \boxed{|\chi\rangle\langle\chi| = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -\sqrt{3} & 0 \\ 0 & -\sqrt{3} & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}.$$

The outer product comes from multiplying this column by $\langle\chi| = \frac{1}{2}[0 \;\; 1 \;\; -\sqrt{3} \;\; 0]$. Its trace is $(1+3)/4 = 1$.

3.3 Why the Bell state is not a product state

Suppose a factorization exists with $|a\rangle = a_0|0\rangle + a_1|1\rangle$ and $|b\rangle = b_0|0\rangle + b_1|1\rangle$. Expanding gives

$$|a\rangle \otimes |b\rangle = a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle.$$

Matching coefficients with $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ requires

$$a_0b_0 = \frac{1}{\sqrt{2}}, \quad a_0b_1 = 0, \quad a_1b_0 = 0, \quad a_1b_1 = \frac{1}{\sqrt{2}}.$$

The first and last equalities imply that all four numbers a_0, a_1, b_0, b_1 are nonzero. Then a_0b_1 and a_1b_0 cannot vanish, a contradiction.

Therefore $|\Phi^+\rangle$ **is not a product state**; it is entangled. Equivalently, every product state satisfies $(a_0b_0)(a_1b_1) = (a_0b_1)(a_1b_0)$, whereas the Bell coefficients would require $1/2 = 0$.

3.4 The matrix of $X \otimes H \otimes Z$

Here X acts on the bottom wire, H on the middle wire, and Z on the top wire. First form the 4×4 matrix acting on the middle and top qubits:

$$B := H \otimes Z = \frac{1}{\sqrt{2}} \begin{bmatrix} Z & Z \\ Z & -Z \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}.$$

Then $X \otimes B = \begin{bmatrix} 0_4 & B \\ B & 0_4 \end{bmatrix}$, where 0_4 is the 4×4 zero matrix. In lexicographic basis order, the full matrix is

$$X \otimes H \otimes Z = \frac{1}{\sqrt{2}} \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \end{array}\right].$$

The block structure reflects the fact that X flips the leftmost bit, on the bottom wire. For example, the first column is

$$(X \otimes H \otimes Z)|000\rangle = |1\rangle \otimes |+\rangle \otimes |0\rangle = \frac{|100\rangle + |110\rangle}{\sqrt{2}},$$

which agrees with the two nonzero entries in that column.

Question 4
Applying tensor-product operators

4.1 Single-qubit actions

Multiplication by X interchanges the entries; multiplication by Z changes the sign of the second entry:

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle, \qquad X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle,$$

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle, \qquad Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle.$$

4.2 The matrix and basis-state action of $X \otimes Z$

Using $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the block form is

$$M := X \otimes Z = \begin{bmatrix} 0_2 & Z \\ Z & 0_2 \end{bmatrix} = \boxed{\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}}.$$

Multiplication by each computational-basis column selects the corresponding column of M:

$$M \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle, \qquad M \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} = -|11\rangle,$$

$$M \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle, \qquad M \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix} = -|01\rangle.$$

For the Dirac-notation calculation, use $(A \otimes B)(|a\rangle \otimes |b\rangle) = (A|a\rangle) \otimes (B|b\rangle)$:

$$\begin{aligned} (X \otimes Z)|00\rangle &= (X|0\rangle) \otimes (Z|0\rangle) = |10\rangle, \\ (X \otimes Z)|01\rangle &= (X|0\rangle) \otimes (Z|1\rangle) = -|11\rangle, \\ (X \otimes Z)|10\rangle &= (X|1\rangle) \otimes (Z|0\rangle) = |00\rangle, \\ (X \otimes Z)|11\rangle &= (X|1\rangle) \otimes (Z|1\rangle) = -|01\rangle. \end{aligned}$$

Both methods give the same four outputs.

4.3 Action on an arbitrary two-qubit state

Let

$$|\phi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle, \qquad |\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1.$$

Dirac notation. Apply the four results of Question 4.2 term by term:

$$\begin{aligned}(X \otimes Z)|\phi\rangle &= \alpha|10\rangle - \beta|11\rangle + \gamma|00\rangle - \delta|01\rangle \\ &= \boxed{\gamma|00\rangle - \delta|01\rangle + \alpha|10\rangle - \beta|11\rangle}.\end{aligned}$$

The second line puts the terms back into computational-basis order.

Column vectors. Multiply by the matrix from Question 4.2:

$$(X \otimes Z)|\phi\rangle = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \boxed{\begin{bmatrix} \gamma \\ -\delta \\ \alpha \\ -\beta \end{bmatrix}}.$$

This agrees with the Dirac expression. Its squared norm is still $|\gamma|^2 + |\delta|^2 + |\alpha|^2 + |\beta|^2 = 1$.

4.4 Connecting the one- and two-qubit descriptions

For bits $a, b \in \{0, 1\}$,

$$X|a\rangle = |a \oplus 1\rangle, \qquad Z|b\rangle = (-1)^b|b\rangle,$$

where $\oplus$ denotes addition modulo two. Consequently,

$$\boxed{(X \otimes Z)|ab\rangle = (-1)^b|(a \oplus 1)b\rangle}.$$

The bottom qubit is flipped by X, and terms whose top qubit is 1 acquire a minus sign from Z. The top bit itself does not change. More generally,

$$(X \otimes Z)(|a\rangle \otimes |b\rangle) = (X|a\rangle) \otimes (Z|b\rangle)$$

for any single-qubit kets $|a\rangle$ and $|b\rangle$. Linearity extends this rule to arbitrary two-qubit superpositions, including entangled inputs.

Why the signs matter. A minus sign on an entire basis state is a global phase if that basis state is the whole input. Within a superposition, different signs can change relative phase. For example,

$$(X \otimes Z)|0+\rangle = |1-\rangle.$$

Thus the phase flip cannot be discarded when applying the operator to a general state.

Question 5
Circuits and gadgets

5.1 SWAP, CNOT and CZ

The columns of a gate matrix are its outputs on the ordered basis states. For the matrices in the question, those outputs are

Input	SWAP	$\mathrm{CNOT}_{1\to0}$	CZ
$\|00\rangle$	$\|00\rangle$	$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|10\rangle$	$\|01\rangle$	$\|01\rangle$
$\|10\rangle$	$\|01\rangle$	$\|11\rangle$	$\|10\rangle$
$\|11\rangle$	$\|11\rangle$	$\|10\rangle$	$-\|11\rangle$

Equivalently, for $a, b \in \{0, 1\}$,

$$\boxed{\begin{aligned} \mathrm{SWAP}|ab\rangle &= |ba\rangle, \\ \mathrm{CNOT}_{1\to0}|ab\rangle &= |a\,(a \oplus b)\rangle, \\ \mathrm{CZ}|ab\rangle &= (-1)^{ab}|ab\rangle. \end{aligned}}$$

SWAP exchanges the qubits. The given CNOT has the bottom wire q_1 as control and the top wire q_0 as target: it flips the rightmost bit exactly when the leftmost bit is 1. CZ changes the sign exactly when both qubits are 1; either qubit can be regarded as its control.

For an arbitrary input with coefficient vector $(\alpha, \beta, \gamma, \delta)^{\mathsf{T}}$, their output columns are, respectively,

$$\begin{bmatrix} \alpha \\ \gamma \\ \beta \\ \delta \end{bmatrix}, \qquad \begin{bmatrix} \alpha \\ \beta \\ \delta \\ \gamma \end{bmatrix}, \qquad \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ -\delta \end{bmatrix}.$$

Comparison with local tensor-product gates. A gate $A \otimes B$ acts independently on the two qubits and maps every product state to a product state: $(A \otimes B)|a\rangle|b\rangle = (A|a\rangle) \otimes (B|b\rangle)$. CNOT and CZ can create entanglement, so neither is of this form. For example,

$$\mathrm{CNOT}_{1\to0}|{+}0\rangle = |\Phi^+\rangle, \qquad \mathrm{CZ}|{+}{+}\rangle = \frac{|00\rangle + |01\rangle + |10\rangle - |11\rangle}{2}.$$

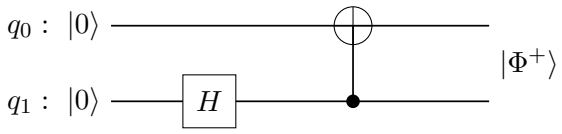
For the CZ output, the product-state condition $c_{00}c_{11} = c_{01}c_{10}$ fails: the two sides are $-1/4$ and $1/4$.

SWAP also cannot equal $A \otimes B$, but it *does* preserve product states: $\mathrm{SWAP}(|a\rangle \otimes |b\rangle) = |b\rangle \otimes |a\rangle$. To see why it cannot be local, compare inputs $|00\rangle$ and $|01\rangle$. A fixed local A would give the same bottom-qubit state $A|0\rangle$ for both inputs, while SWAP gives bottom-qubit states $|0\rangle$ and $|1\rangle$. Thus being nonlocal does not by itself imply that a gate can entangle a product input.

5.2 Preparing $|\Phi^+\rangle$ from $|00\rangle$

Apply a Hadamard to the bottom qubit q_1, followed by a CNOT with the bottom qubit as control and the top qubit q_0 as target:

$$\boxed{|\Phi^+\rangle = \mathrm{CNOT}_{1\to 0}(H \otimes I)|00\rangle}.$$



The state after the Hadamard is

$$(H \otimes I)|00\rangle = |+\rangle \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

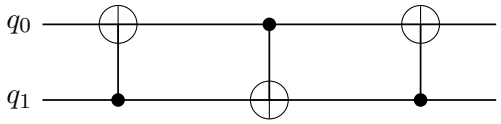
The CNOT leaves $|00\rangle$ unchanged and maps $|10\rangle$ to $|11\rangle$, producing $(|00\rangle + |11\rangle)/\sqrt{2}$. No measurement is required for this preparation.

5.3 Constructing SWAP from CNOTs

Use three CNOTs, with control directions bottom-to-top, top-to-bottom, and bottom-to-top:

$$\boxed{\mathrm{SWAP} = \mathrm{CNOT}_{1\to 0}\ \mathrm{CNOT}_{0\to 1}\ \mathrm{CNOT}_{1\to 0}}.$$

The rightmost CNOT acts first. No single-qubit gates are needed when both CNOT directions are available.



On an arbitrary basis state $|a, b\rangle$, where a is the bottom bit and b the top bit, the successive bit values are

$$\begin{aligned} |a, b\rangle &\xrightarrow{\mathrm{CNOT}_{1\to 0}} |a, a \oplus b\rangle \\ &\xrightarrow{\mathrm{CNOT}_{0\to 1}} |a \oplus (a \oplus b), a \oplus b\rangle = |b, a \oplus b\rangle \\ &\xrightarrow{\mathrm{CNOT}_{1\to 0}} |b, (a \oplus b) \oplus b\rangle = |b, a\rangle. \end{aligned}$$

This uses $a \oplus a = b \oplus b = 0$. Since basis states span the two-qubit space, the circuit swaps arbitrary superpositions as well. The explicit matrix calculation follows.

5.3 (continued) Explicit matrix verification

In the order $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$, define

$$C := \mathrm{CNOT}_{1\to 0} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D := \mathrm{CNOT}_{0\to 1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

The circuit matrix is CDC. First multiply the matrices for the first two gates in time order:

$$DC = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Now multiply by the final CNOT on the left:

$$C(DC) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \boxed{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}} = \mathrm{SWAP}\,.$$

This is exactly the matrix in the question, including all phases.

If only the given CNOT direction is used. Single-qubit Hadamards reverse the direction:

$$\boxed{D = (H \otimes H)C(H \otimes H)}.$$

For completeness, let $K = H \otimes H = \frac{1}{2}M$, where

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

Multiplication gives

$$KCK = \frac{1}{4}MCM = \frac{1}{4}\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 4 & 0 \\ 0 & 4 & 0 & 0 \end{bmatrix} = D.$$

Substituting into CDC yields a construction using only the CNOT matrix supplied in the problem and single-qubit gates:

$$\boxed{\mathrm{SWAP} = C(H \otimes H)C(H \otimes H)C}.$$

In time order: apply C, then H to both qubits, then C, then H to both qubits, and finally C.